

Supplementary Material

APPENDIX D PROOF OF LEMMA 1

With the definitions of w in (28) and $\mathcal{W}_\lambda$ in (31), $w \in \mathcal{W}_\lambda$ if and only if

$$\begin{aligned} x &\in \mathcal{X}, & \chi^r &\in \mathcal{X}_\lambda^r, & \nu^r &\in \mathcal{U}_\lambda^r, \\ \kappa(w) &\in \mathcal{U}, & (A_T - P_X)\chi^r + B_T\nu^r &= 0. \end{aligned}$$

Using the generator representations of the four constraint sets, these conditions hold if and only if there exist $\xi_X, \xi_{X^r}, \xi_{U^r}, \xi_U$, satisfying $\|\xi_X\|_\infty \leq 1, \|\xi_{X^r}\|_\infty \leq 1, \|\xi_{U^r}\|_\infty \leq 1, \|\xi_U\|_\infty \leq 1$, such that

$$x = c_X + G_X \xi_X, \quad (\text{D.1})$$

$$\chi^r = c_{X^r} + \lambda G_{X^r} \xi_{X^r}, \quad (\text{D.2})$$

$$\nu^r = c_{U^r} + \lambda G_{U^r} \xi_{U^r}, \quad (\text{D.3})$$

$$\kappa(w) = c_U + G_U \xi_U, \quad (\text{D.4})$$

$$0 = (A_T - P_X)\chi^r + B_T\nu^r. \quad (\text{D.5})$$

Substituting (D.1)–(D.3) into (D.4) and using (29) yields

$$\begin{aligned} &-KG_X \xi_X + \lambda KE_x G_{X^r} \xi_{X^r} + \lambda E_u G_{U^r} \xi_{U^r} - G_U \xi_U \\ &= c_U + Kc_X - KE_x c_{X^r} - E_u c_{U^r}. \end{aligned} \quad (\text{D.6})$$

Similarly, substituting (D.2) and (D.3) into (D.5) gives

$$\begin{aligned} &\lambda(A_T - P_X)G_{X^r} \xi_{X^r} + \lambda B_T G_{U^r} \xi_{U^r} \\ &= -(A_T - P_X)c_{X^r} - B_T c_{U^r}. \end{aligned} \quad (\text{D.7})$$

Collect the generator coefficients into

$$\xi_{\mathcal{W}} = [\xi_X^\top \quad \xi_{X^r}^\top \quad \xi_{U^r}^\top \quad \xi_U^\top]^\top.$$

By the definitions of $G_{\mathcal{W}}$ and $c_{\mathcal{W}}$ in (36a) and (36b), equations (D.1)–(D.3) are equivalent to $w = c_{\mathcal{W}} + G_{\mathcal{W}}\xi_{\mathcal{W}}$, and, by the definitions of $A_{\mathcal{W}}$ and $b_{\mathcal{W}}$ in (36c) and (36d), (D.6) and (D.7) are equivalent to $A_{\mathcal{W}}\xi_{\mathcal{W}} = b_{\mathcal{W}}$. Also, the individual bounds on the generator coefficients are equivalent to $\|\xi_{\mathcal{W}}\|_\infty \leq 1$. Hence

$$w \in \mathcal{W}_\lambda \iff \exists \xi_{\mathcal{W}} : \begin{cases} w = c_{\mathcal{W}} + G_{\mathcal{W}}\xi_{\mathcal{W}}, \\ A_{\mathcal{W}}\xi_{\mathcal{W}} = b_{\mathcal{W}}, \\ \|\xi_{\mathcal{W}}\|_\infty \leq 1. \end{cases}$$

The generator coefficient ξ_U does not affect the parameterization of w , as reflected by the corresponding zero block in $G_{\mathcal{W}}$. Its existence, together with (D.6), is instead equivalent to $\kappa(w) \in \mathcal{U}$. Therefore, by the definition of a constrained zonotope in (2),

$$\mathcal{W}_\lambda = \langle G_{\mathcal{W}}, c_{\mathcal{W}}, A_{\mathcal{W}}, b_{\mathcal{W}} \rangle$$

thus concluding the proof.

APPENDIX E PROOF OF LEMMA 3

The case $q = 1$ is established in Lemma 2. Assume now that the result holds for Ω_{q-1} , with $q \geq 2$. By (38),

$$\Omega_q = \Omega_{q-1} \cap \text{Pre}(\Omega_{q-1}).$$

Therefore,

$$w \in \Omega_q \iff w \in \Omega_{q-1}, \quad \Phi_w w \in \Omega_{q-1}.$$

According to the assumed representation, membership in Ω_{q-1} is characterized exactly by

$$\begin{aligned} w_{q-1} &= G_{q-1}^{\text{red}} \zeta_{q-1}^{\text{red}} + c_{\mathcal{W}}, \\ A_{q-1}^{\text{red}} \zeta_{q-1}^{\text{red}} &= b_{q-1}^{\text{red}}, \quad \|\zeta_{q-1}^{\text{red}}\|_\infty \leq 1. \end{aligned}$$

Since each pair of consecutive successors is related by the same one-step dynamics considered in Lemma 2, the reduction established therein can be applied at every step. In particular, shifting the reference sequence (χ_j^r, ν_j^r) is equivalent to shifting the corresponding generator coefficients $(\xi_{X^r,j}, \xi_{U^r,j})$. As such, the reference coefficients can be selected, without loss of exactness, as

$$\xi_{X^r,j} = P_{\xi_X} \xi_{X^r,j-1}, \quad \xi_{U^r,j} = P_{\xi_U} \xi_{U^r,j-1},$$

which reduces to (B.8) when $j = 1$. Repeated iterations therefore yield

$$\xi_{X^r,j} = P_{\xi_X}^j \xi_{X^r,0}, \quad \xi_{U^r,j} = P_{\xi_U}^j \xi_{U^r,0}, \quad j \in \mathbb{I}_{[1,q]}. \quad (\text{E.1})$$

As shown in Lemma 2, these selections preserve the corresponding component-wise infinity-norm bounds and satisfy the periodic-reference-dynamics equalities in (27). Hence, at the q -th successor, the only new coefficients that must be retained are $\eta_q = [\xi_{X,q}^\top \quad \xi_{U,q}^\top]^\top$. Substituting (E.1) with $j = q$ into the auxiliary-input admissibility constraint $\kappa(w_q) \in \mathcal{U}$ at the q -th successor gives

$$\begin{aligned} &\lambda KE_x G_{X^r} P_{\xi_X}^q \xi_{X^r,0} + \lambda E_u G_{U^r} P_{\xi_U}^q \xi_{U^r,0} \\ &- KG_X \xi_{X,q} - G_U \xi_{U,q} = b^{(a)}. \end{aligned}$$

By (50a) and (51a), this equality can be written as

$$A_q^{(a)} \xi_0 + \Delta^{(a)} \eta_q = b^{(a)}. \quad (\text{E.2})$$

Similarly, the physical-state transition in (25) from the $(q-1)$ -th to the q -th successor depends on the artificial reference at step $q-1$. Using (E.1) gives

$$\begin{aligned} &(A - BK)G_X \xi_{X,q-1} + \lambda BKE_x G_{X^r} P_{\xi_X}^{q-1} \xi_{X^r,0} \\ &+ \lambda BE_u G_{U^r} P_{\xi_U}^{q-1} \xi_{U^r,0} - G_X \xi_{X,q} = b^{(b)}. \end{aligned}$$

By (50b), (51b), and (51c), this equality is equivalently written as

$$A_q^{(b)} \xi_0 + \Delta^{(c)} \eta_{q-1} + \Delta^{(b)} \eta_q = b^{(b)}. \quad (\text{E.3})$$

Therefore, the exact representation of Ω_q is obtained from that of Ω_{q-1} by appending η_q to the coefficient vector and adding (E.2) and (E.3). Among the block columns associated with $\eta_1, \dots, \eta_q$, each input-admissibility equality contains $\Delta^{(a)}$ only in the block column corresponding to η_j . The first state-transition equality contains $\Delta^{(b)}$ in the block column corresponding to η_1 , whereas, for $j \geq 2$, the corresponding equality contains $\Delta^{(c)}$ and $\Delta^{(b)}$ in the block columns corresponding to η_{j-1} and η_j , respectively.

Collecting first the q input-admissibility equalities of the form (E.2) and then the q state-transition equalities of the form (E.3) amounts only to a permutation of the equality

constraints and yields exactly the block structure in (52). The same permutation of the corresponding right-hand sides gives (53).

By the assumed representation of Ω_{q-1} , the columns of G_{q-1}^{red} associated with $\eta_1, \dots, \eta_{q-1}$ are already zero. The newly introduced coefficients in η_q are required to enforce the equality constraints associated with state and input admissibility at the q -th successor. However, they do not contribute to the generator equation mapping the constrained-zonotope coefficients to w . Consequently, appending η_q adds zero columns to G_{q-1}^{red} , yielding (49). Moreover, since ζ_q^{red} is obtained by appending η_q to ζ_{q-1}^{red} , $\|\zeta_q^{\text{red}}\|_{\infty} \leq 1$ adds exactly the component-wise bounds on η_q to those already contained in the representation of Ω_{q-1} . Thus, the reduced representation remains exact, and the result follows by induction.